

LoRaWAN Sub-Surface Precision Network: Detailed Report

Abstract

The LoRaWAN Sub-Surface Precision Network proposes a fully buried sensor network for precision agriculture (zero-footprint, 'root-eye' sensing) that uses LoRa/LoRaWAN radio to send soil data from multiple depths to above-ground gateways. The system combines sub-surface sensing (capacitive/TDR/SIS), soil-matched antenna and enclosure design, adaptive RF (moisture-aware spreading factor / link control), and farmer-friendly maintenance (acoustic/NFC homing, biodegradable housings). The document outlines prototype hardware choices (ESP32-LoRa, PVC enclosures, capacitive sensors), experiments (mud-bucket tests, depth profiling), and patentable innovations (soil-matched antenna, moisture-adaptive protocol, acoustic locator).

What you are doing (one-liner)

Designing a robust, maintainable, low-cost buried LoRaWAN sensor network that survives farm operations, delivers high-fidelity multi-depth soil data, and provides novel technical/IP contributions on the underground-to-aboveground (UG2AG) link.

Short verdict — feasibility & viability

Technical feasibility (student prototype): High. A two-node mud-bucket proof and multi-depth prototype with an above-ground gateway are realistic.

Field-scale feasibility: Challenging but doable due to RF attenuation in wet/high-permittivity soils; expect tens to a few hundred meters depending on depth and moisture.

Commercial viability in India: Conditional — viable if cost per node is reduced and maintenance/localization challenges are solved.

Detailed technical analysis

1) Physics & propagation (core constraint)

Soil behaves as a lossy dielectric and RF attenuation grows rapidly with volumetric water content (VWC) and frequency. Sub-GHz LoRa helps, but path loss in soil is orders of magnitude worse than in air. Empirical testing (bucket → field) is essential to quantify site-specific path loss. Literature indicates practical UG2AG transmission ranges for RF systems of roughly 4–20 m in many conditions, though results vary by soil type and moisture levels.

2) Sensing layer

Sensors: capacitive probes (low cost) vs Time-Domain Reflectometry (TDR, accurate but more expensive) vs Soil Impedance Spectroscopy (SIS) for nutrient signatures. Recommendation: use capacitive probes for initial prototypes and calibrate against a TDR reference per soil type.

3) Antenna & enclosure

Buried antennas experience severe detuning. Design a soil-matched PCB antenna and/or integrate a reflective ground-plane under the node to direct energy upward. Installation must avoid air gaps; use soil slurry packing or screw-in installations. Consider a shallow antenna chimney if allowable.

4) Link robustness & protocol

Implement moisture-aware adaptive LoRa parameters: nodes estimate local VWC and link metrics and proactively adjust Spreading Factor (SF) and transmit power (e.g., SF7→SF12 when attenuation increases). Collect field data to design an adaptive controller that balances energy use and reliability.

5) Power & longevity

Options include primary Li-SOCl₂ cells with deep sleep for multi-year life, small solar surface collectors, or high-research inductive top-up methods (e.g., tractor-mounted coil). Pragmatic approach: deep sleep + infrequent TX for initial prototypes.

6) Maintenance & localization

Locating failed nodes is a core maintenance challenge. Possible solutions: NFC/HF passive markers, magnet + detector, low-power acoustic/seismic pinger, or surface RFID. Acoustic homing is promising and novel; evaluate signal propagation through soil and practical handheld detection ranges.

7) Topology & scaling

Star topology with above-ground gateways is currently most practical. Gateways mounted on poles across the farm provide coverage; design density based on attenuation measurements.

Concrete experiments (week → quarter plan)

Week 1–3 (mud-bucket): One ESP32-LoRa node in a bucket at 30 cm depth. Measure RSSI/SNR for dry to wet conditions and log VWC vs path loss. **Month 2–4 (depth profile):** Deploy nodes at 20/50/100 cm; gateway at 50–200 m. Build attenuation maps and tune antenna. **Month 4–8 (field pilot):** 5 nodes across 1–2 acres through monsoon. Test adaptive SF algorithm. **Deliverables:** Dataset (VWC vs path loss), adaptive SF algorithm, PCB antenna design, BOM, provisional patent draft.

Patent research in India — summary & recommendations

Existing prior art includes underground sensing patents and LoRa+soil sensor inventions. Avoid overly broad claims on 'buried sensor measures moisture' — instead claim novel mechanisms such as antenna geometry, moisture-aware protocol, slurry installation, biodegradable enclosure, acoustic homing, or inductive charging through soil. File a provisional application in India for the strongest inventive concept and collect lab logs, datasets, CADs, and RF plots as evidence.

Related research topics you can publish on

1. UG2AG attenuation models for Indian soil types (black cotton vs alluvial). 2. Adaptive LoRaWAN control: moisture-aware SF/power algorithms. 3. Soil Impedance Spectroscopy (SIS) for in-situ NPK estimation. 4. Soil-matched antenna design and VNA characterization. 5. Wireless power transfer through soil and land-vehicle based top-ups. 6. Maintenance & localization tradeoffs: acoustic/seismic homing vs RFID.

Literature Review (selected recent papers)

Wireless Underground Sensor Communication Using LoRa: challenges and propagation (MDPI, 2024)

Comprehensive study of underground RF propagation, noting low transmission ranges (4–20 m typically) and recommending hop networks, antenna design and soil-specific calibration. See: <https://www.mdpi.com/1424-8220/24/10/3113>

Application of Wireless Underground Sensor Networks (MDPI, 2023)

Describes hop network approaches to overcome UG2AG distance limitations and shows deployment case studies. See: <https://www.mdpi.com/1424-8220/23/8/3795>

Time-Domain Reflectometry (TDR) for soil moisture (Ledieu, 1986)

Foundational TDR method for measuring soil moisture by permittivity measurement; highly accurate baseline for calibration.

<https://www.sciencedirect.com/science/article/pii/0022169486900971>

Wireless Power Transfer Through Soil (NSF/Research, 2021)

Studies feasibility and limitations of inductively coupled WPT through soil across moisture levels; shows limited but possible power transfer at shallow depths.

<https://par.nsf.gov/servlets/purl/10313152>

SMoL: Sensing Soil Moisture Using LoRa (ACM, 2022)

Demonstrates low-cost soil moisture estimation using LoRa RSSI as an additional sensing modality; useful for low-cost deployments. <https://dl.acm.org/doi/10.1145/3477085.3478991>

References

Al Moshi, M.A., 'Wireless Underground Sensor Communication Using LoRa', Sensors (MDPI), 2024. <https://www.mdpi.com/1424-8220/24/10/3113>

Liang, M.C., 'Application of Wireless Underground Sensor Networks', Sensors (MDPI), 2023. <https://www.mdpi.com/1424-8220/23/8/3795>

Ledieu, J., 'A method of measuring soil moisture by time-domain reflectometry', 1986. <https://www.sciencedirect.com/science/article/pii/0022169486900971>

Luo, W., 'Wireless Power Transfer Through Soil', NSF/Research, 2021. <https://par.nsf.gov/servlets/purl/10313152>

Kiv, D., 'SMoL: Sensing Soil Moisture using LoRa', ACM, 2022. <https://dl.acm.org/doi/10.1145/3477085.3478991>

Additional references and patent sources: Google Patents, Indian Patent Advanced Search System (InPASS).

Rough BOM & cost (prototype)

Name of Equipment	Estimated Cost (INR)
ESP32-LoRa dev board	■1,500–1,800
Capacitive soil probe	■150–■500
Battery (18650 / Li-SOCl ₂)	■250
PVC pipe + epoxy	■200–■400
Misc (antenna PCB, potting compound)	■300

Immediate recommended next steps

- Run the mud-bucket test (Week 1) and collect RSSI/SNR vs VWC. - Build attenuation curves for local soils. - Prototype antenna with small PCB reflector and test in bucket. - Prepare provisional patent materials for strongest inventive concept.